



The Pythagorean Theorem

A right triangle has a **right angle**, which equals 90°. The two sides that form the right angle are called **legs**, and the side opposite the right angle is the **hypotenuse**. The hypotenuse is always the longest side of a right triangle.

The relationship between the legs and hypotenuse of a right triangle was discovered almost 2,500 years ago by a Greek mathematician named Pythagoras.

Pythagorean theorem $a^2 + b^2 = c^2$

The sum of the squares of the legs of a right triangle equals the square of the hypotenuse.

Example 1

[Hide](#)

The lengths of the legs of a right triangle are 9 and 12. What is the hypotenuse?

Step 1

Substitute 9 for a and 12 for b in the Pythagorean theorem formula.

$$a^2 + b^2 = c^2$$
$$9^2 + 12^2 = c^2$$

Step 2

Evaluate the left side of the equation.

$$\sqrt{81 + 144} = c$$
$$\sqrt{225} = c$$

Step 3

Find c .

$$15 = c$$

The hypotenuse is **15 units** long.

Example 2

[Hide](#)

The hypotenuse of a right triangle has a length of 10 units, and one of the legs is 6 units long. Find the measure of the other leg.

Step 1

Substitute 10 for c and 6 for a in the Pythagorean theorem formula.

$$a^2 + b^2 = c^2$$
$$6^2 + b^2 = 10^2$$

Step 2

Simplify the equation.

$$36 + b^2 = 100$$
$$b = \sqrt{64}$$

Step 3

Find b .

$$b = 8$$

The other leg is **8 units** long.

Key Terms

 **hypotenuse**

 **Pythagorean theorem**

 **right angle**

Rate Your Confidence

 High

 Medium

 Low

Continue >>

Category

Lesson 6: Problem Solving with Triangles

Related Questions:

[Practice questions taken related to this topic](#)

 Report Content Errors